

ECON 21030: Honors Econometrics

Homework 1

Spring Quarter 2024

Question 1: [10 points] The *moment generating function* (MGF) of a distribution is defined as $M_X(t) = E[\exp(tX)]$, whenever this expectation exists in a neighborhood of $t = 0$.

1. Write down and simplify as much as possible the MGF for a normal distribution with mean μ and variance σ^2 . Show your work.
2. What are the first and second derivatives of the MGF for this normal random variable at $t = 0$? How do they relate to its *moments*?
3. Now take a third-order Taylor expansion of $\exp(tX)$ around zero. What is its expectation? What does this lead you to believe about moment generating functions more generally?
4. Consider an iid sequence of random variables $\{X_1, \dots, X_n\}$ with finite mean μ and finite variance σ^2 . Define $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{X_i - \mu}{\sigma}$, i.e. the standardized sample mean. How does the MGF of Z_n relate to the MGF of $Y_i = \frac{X_i - \mu}{\sigma}$?
5. Write down the second-order expansion of $M_Y(t)$ around zero.
6. Using this approximation, evaluate the limit of $M_{Z_n}(t)$ as $n \rightarrow \infty$.
7. What distribution does this limit remind you of? What have you proven through this exercise?

Question 2: [10 points] Consider an iid sample $\{X_1, \dots, X_N\}$ from a normal distribution with mean μ and variance σ^2 .

1. Write down the usual estimators of μ and σ^2 . Show that they are unbiased.
2. Compute a 95% confidence interval for μ as a function of these estimators and N . How does its length change with N ?
3. What is the sampling distribution of $\hat{\sigma}^2$? You may use the fact that $\sum_{i=1}^N (X_i - \bar{X})/\sigma^2 \sim \chi^2(N-1)$. Explain how to compute a 95% confidence interval for σ^2 as a function of this distribution, $\hat{\sigma}^2$, and N .
4. Assuming $\sigma^2 = 1$, plot the distribution of the sampling variance of $\hat{\sigma}^2$ from part three for $N \in \{10, 50, 100\}$. What do you notice? How does it compare to a normal with mean σ^2 and variance $\frac{2\sigma^4}{N-1}$? Make a quantile-quantile plot to compare for each N .
5. Provide an alternative (and simpler) confidence interval for $\hat{\sigma}^2$ based on this observation. How does the length of this confidence interval scale with N ?

- Now use the same approximation to construct a confidence interval for your *standard error* for $\bar{X}$. You will need to apply the delta method, which says that if $X \rightarrow^D N(\mu, V)$, $f(X) \rightarrow^D N(f(\mu), V f'(\mu)^2)$. How does its length scale with N ? What does that tell you about how much easier it is to estimate the standard error of the sample mean than the mean itself?

Question 3: [10 points] Suppose we have an iid sample $\{X_i\}_{i=1}^N$ of uniform random variables with support $[0, \theta]$.

- Consider the maximum order statistic $X_{max} = \max\{X_1, \dots, X_N\}$. Is X_{max} an unbiased estimator of θ ? Why or why not?
- Is X_{max} a consistent estimator of θ ? Why or why not?
- Consider the $n-1$ order statistic, i.e., the second largest sample value. You can call it $X_{(n-1)}$. Is it an unbiased and/or consistent estimator of θ ? How does its bias (if any) compare to the bias of the maximum order statistic?
- Propose a different, unbiased estimator of θ based on the sample mean. How does its variance compare to the maximum order statistic?
- Which estimator would you prefer and why? Can you think of an adjustment for the maximum order statistic that might improve any bias it might have? What is the effect on its variance? Is there any reason to use the sample average after making that adjustment?
- Explain what you have learned about choice of estimators, bias, and variance.

Question 4: [10 points] A common way to evaluate the performance of estimators is its *mean-square* error, defined as:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

- Show that the MSE can be decomposed as:

$$MSE(\hat{\theta}) = (E[\hat{\theta}] - \theta)^2 + Var(\hat{\theta})$$

Discuss these two terms, and why MSE would be a reasonable way to evaluate an estimator.

- Write some code to simulate the MSE of the sample max, its unbiased adjustment, and the sample mean estimators of θ considered in the previous problem. Your code should fix θ (e.g., at 1), set $n = 10$, and repeatedly draw random samples. In each sample, compute the estimators and evaluate squared error. Average over many such draws—enough so that your simulation is stable.
- How does MSE break down into bias and variance for each estimator, and how does total MSE and its composition vary as a function of θ ? Make a graph with clearly labeled axes and series communicating the results.
- Now consider a new estimator of θ defined as $\hat{\theta}(\lambda) = \lambda X_{max}$. For $n = 3$, what choice of λ would minimize MSE for $\theta = 1$? What about for $\theta = 10$? What about for $n = 10$?
- Provide an analytic interpretation for the optimal choice of lambda. How does it compare to the choice of lambda that makes $\hat{\theta}$ unbiased? How does that make you feel about using unadjusted X_{max} as an estimator?